

MODULE 5.1 · Concept 1 of 8

# What is an AI Agent?

*Defining the building block of autonomous AI — from chatbots to Devin*

**8 min**

Duration

**B2 – Understand**

Bloom's

**L2 – Intermediate**

Level

**M1–M4**

Prerequisite

AI-AG-FN-TH-000001

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## SECTION 1 — WHY

# Why Do We Need AI Agents? ?

*The limitations of standalone LLMs that agents solve*

# Limitation 1: LLMs Are Stateless

*Every prompt starts from zero — no memory of what came before*

## Turn 1

User: "What flights go BLR→BOM on March 15?"

LLM: "IndiGo, Air India, SpiceJet all fly that route.  
IndiGo 6:30 AM is cheapest at ₹4,200."

✓ Useful answer — but the conversation ends here.

User:

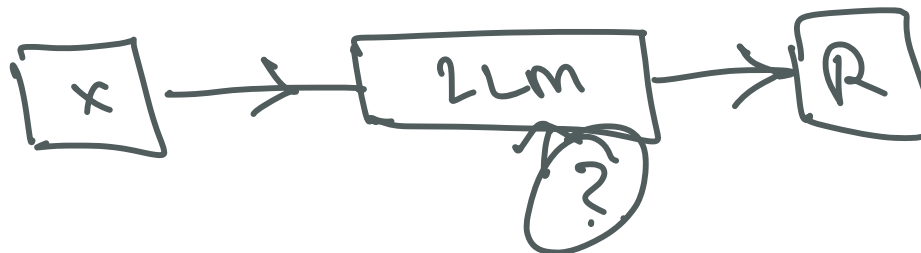
## Turn 2 (new prompt)

User: "Book the IndiGo one"

LLM: "I'm sorry, I don't know which flight  
you're referring to. Could you provide  
more details?"

✗ Turn 1 is completely forgotten. No state carried.

*The LLM has no mechanism to carry information between separate prompts. Each call is independent.*



API call → Indigo? R ← Air India?

## Limitations 2 & 3: No Tools, Single Turn

*LLMs cannot act in the world or iterate toward a solution*

### No Tools

Cannot browse the web, query databases, execute code, call APIs, or access any external system.

Knowledge is frozen at training time. Cannot look up today's flight prices, check weather, or verify facts.

### Single Turn

Generates one response per prompt. No iterative loop.

Cannot:

- Retry a failed approach
- Refine based on intermediate results
- Chain multiple steps toward a goal
- Adapt strategy when first attempt doesn't work

**What these three limitations mean together:** An LLM can answer questions, but it cannot DO things. It cannot check if a flight exists, book it, handle payment failures, or retry with different dates. It generates text — it doesn't solve problems that require real-world interaction.

# Practical: ChatGPT vs AI Agent

The same task — fundamentally different outcomes

Task: "Book me a flight BLR→BOM under ₹5,000 for March 15"

## ChatGPT Response

"I cannot access external websites or booking systems. Here are some general suggestions:

- Try MakeMyTrip or Cleartrip
- IndiGo often has low fares
- Book 2–3 weeks in advance"

→ No action taken. User does all the work.

## AI Agent Response

Thought: I need to search flights

Action: search\_flights("BLR→BOM", "Mar 15")

Obs: IndiGo ₹4,200 at 6:30 AM

Thought: Under budget. Book it.

Action: book\_flight(IN-4321)

Obs: Confirmed XK9J2

→ "Booked! IndiGo 6:30 AM, ₹4,200.  
Confirmation: XK9J2"

The agent perceives, reasons, acts, and iterates. The chatbot just generates text.



1. Pre-training ✓

2. mid-training  
(appl. focused)

SFT | IFT | Alignment  
[tool capabilities]

## SECTION 2 — WHAT

# Defining an AI Agent

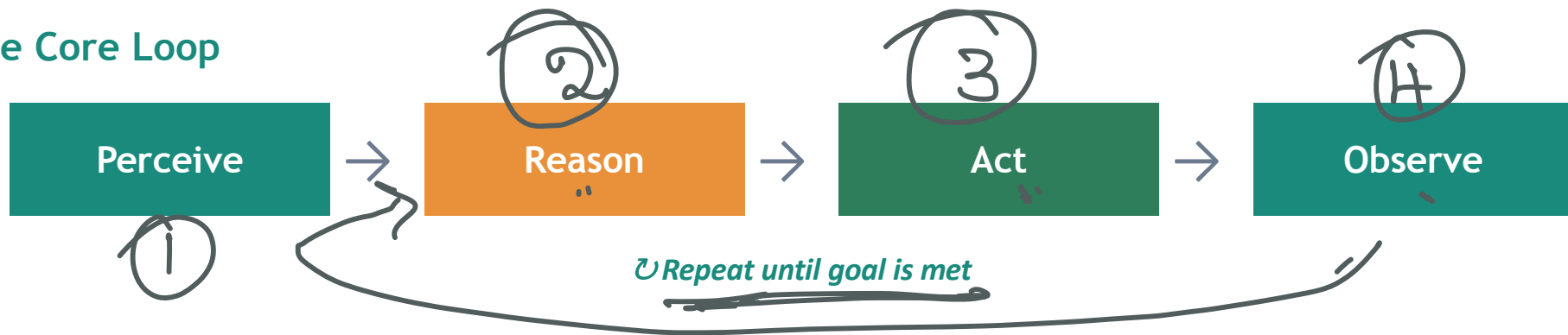
*The formal definition and the five capabilities that make it work*

# The Formal Definition of an AI Agent

A software system with an LLM brain that perceives, reasons, and acts in a goal-directed loop

**An AI Agent** is a system that uses an LLM as its reasoning engine to (1) perceive inputs, (2) maintain state across steps, (3) reason about goals, (4) take actions via tools, and (5) iterate autonomously until the task is complete.

## The Core Loop

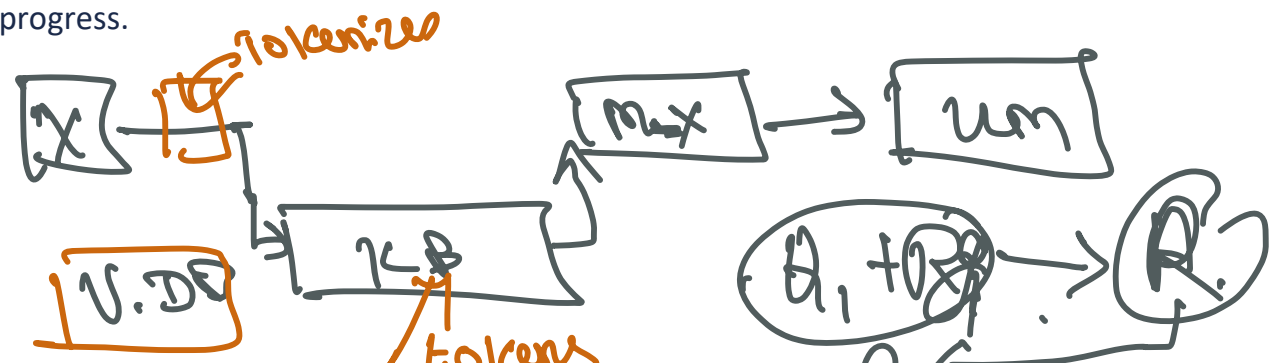
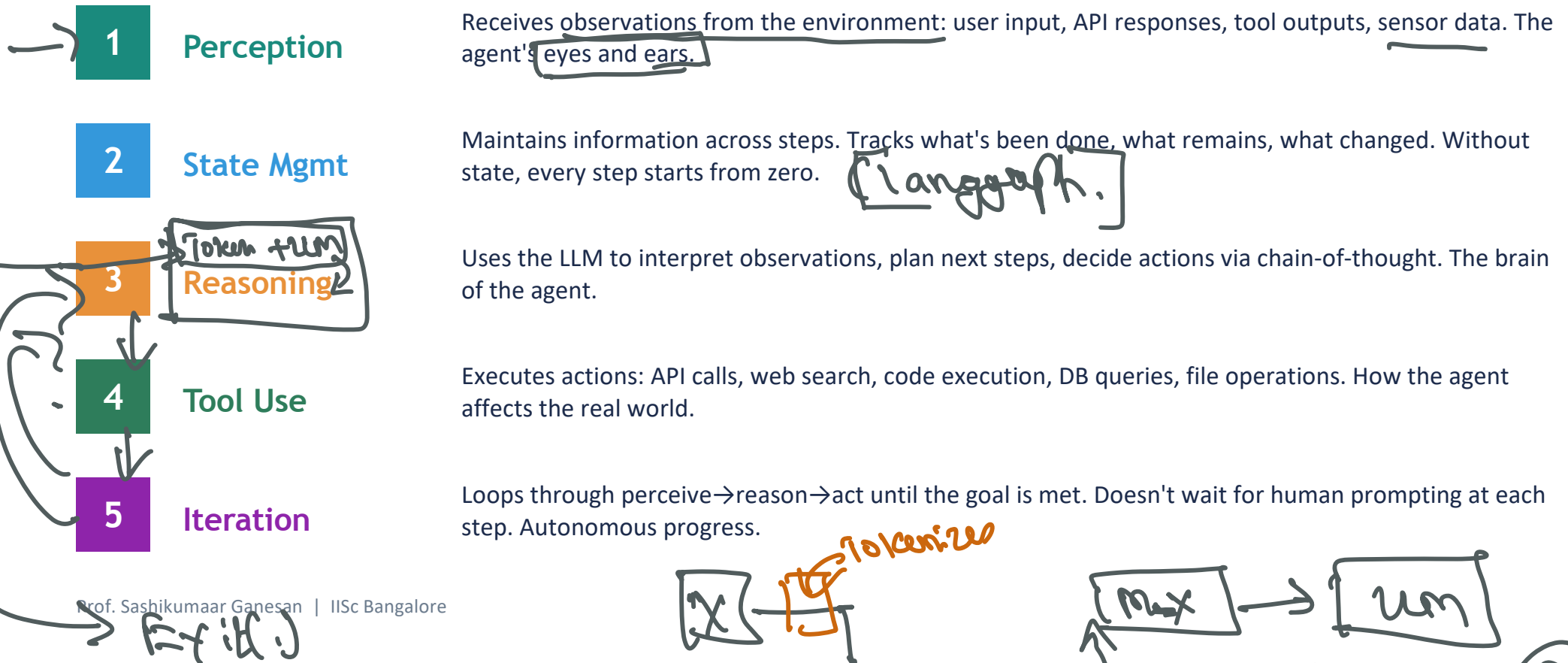


**Key distinction:** A chatbot responds. An agent solves. The loop is what makes the difference.

→ Expert systems (only rule-based)  
→ AI Agent (uses LLM as reasoning engine)

# Five Capabilities That Define an Agent

Remove any one and the system degrades to a chatbot



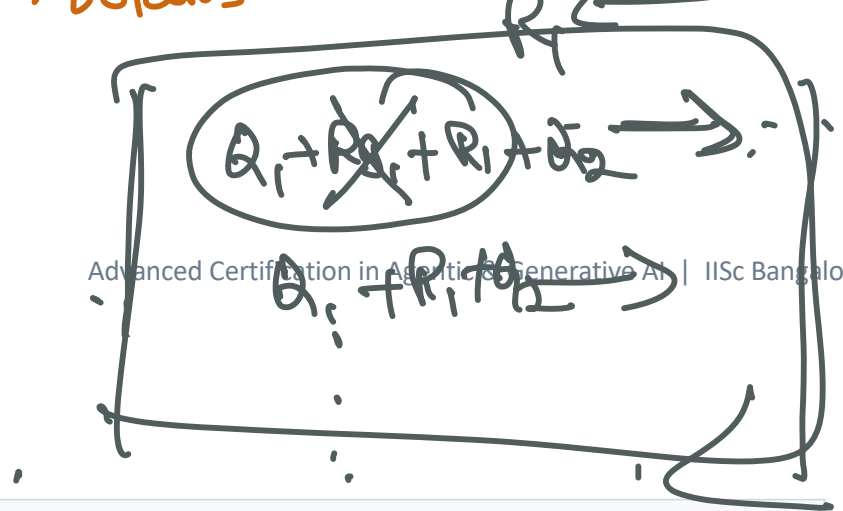


# The Agent Complexity Spectrum

From raw LLMs to multi-agent systems — five levels of capability

|         |                |                      |                                                                                            |
|---------|----------------|----------------------|--------------------------------------------------------------------------------------------|
| Level 0 | Raw LLM        | ChatGPT basic        | Single prompt, single response. No tools, no memory, no loop.                              |
| Level 1 | LLM + Tools    | ChatGPT with plugins | Can call external tools (search, code). Still single-turn — no iterative reasoning.        |
| Level 2 | ReAct Agent    | LangChain agent      | Thought→Action→Observation loop. Reasons about when and which tools to call. Multi-step.   |
| Level 3 | Planning Agent | AutoGPT, BabyAGI     | Decomposes complex goals into sub-tasks. Plans before acting. Can replan on failure.       |
| Level 4 | Multi-Agent    | Devin, Claude Code   | Multiple specialized agents coordinating. Manager delegates to specialists. Full autonomy. |

*Each level adds a capability layer. This module takes you from Level 0 understanding to Level 3–4 building.*

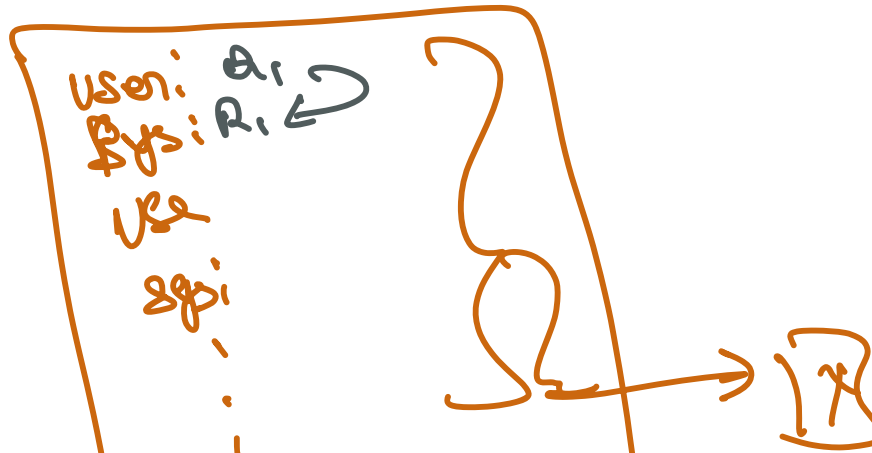


# Chatbot vs. Agent – Six Key Differences

*Why a chatbot is NOT an agent*

| Dimension      | Chatbot                  | AI Agent                       |
|----------------|--------------------------|--------------------------------|
| Turns          | Single turn response     | Multi-step iterative loop      |
| Tools          | None — text in, text out | Web, DB, APIs, code execution  |
| Memory         | Context window only      | Short-term + long-term storage |
| Goal Handling  | Responds to prompt       | Pursues goals autonomously     |
| Error Recovery | User must re-prompt      | Self-corrects and retries      |
| Autonomy       | Human drives every step  | Operates independently         |

*The table makes it clear: the agent has every capability the chatbot lacks. The cost is complexity — agents are harder to build, test, and debug.*



## SECTION 3 — WHERE

# Real-World AI Agents

*Agents are already deployed at scale — here are four production examples*

# Devin: Autonomous Software Engineer

*Cognition Labs — the first AI agent that can plan, code, test, and deploy*

## What Devin Does

- Reads a GitHub issue or task description (Perception)
- Plans multi-file architecture changes (Reasoning)
- Writes code across multiple files simultaneously (Action)
- Runs tests, reads error output (Observation)
- Fixes bugs and iterates until tests pass (Loop)
- Creates pull requests with documentation



## Agent Anatomy Mapping

Perception: Issue description + codebase AST

Memory: File index + recent edits + test results

Reasoning: GPT-4 plans multi-file changes

Action: Code editor + terminal + browser

Iteration: Loops until all tests pass

Level: 4 (Multi-Agent)

Cost: ~\$0.50–\$5.00 per task resolution

# Klarna AI: Customer Support at Scale

*2/3 of all customer chats handled by AI — equivalent to 700 full-time agents*

**2/3**

of all chats  
handled by AI

**700**

full-time agents  
replaced

**\$40M**

annual savings  
projected

**< \$0.001**

cost per  
resolution

## Cost Calculation — Why AI Agents Transform Support Economics

### Pre-AI:

700 agents × \$8/hour × 2,000 hours/year = \$11.2M/year in labor costs alone.

**Post-AI:** 50K tickets/month × \$0.0003/resolution = \$15/month = \$180/year for the AI portion.

The AI doesn't eliminate all human agents — complex cases still escalate. But it handles the routine 2/3 at near-zero marginal cost.

# Elicit & Claude Code: Research and Coding Agents

*Two more production agents operating at different points on the spectrum*

## Elicit — Research Agent (Level 3)

What it does:

- Searches academic paper databases
- Extracts key findings from each paper
- Synthesizes claims across multiple sources
- Generates structured summaries with citations

Why Level 3: Plans which papers to read based on query analysis. Decomposes research into search → read → extract → synthesize steps.

## Claude Code — Terminal Agent (Level 4)

What it does:

- Reads entire codebases via terminal
- Implements features across multiple files
- Runs tests and reads error output
- Iterates on fixes until tests pass

Why Level 4: Uses multiple sub-agents for different tasks (file search, code writing, test execution). Full terminal access = real-world actions.

***Notice the pattern: every production agent maps to the same four components — just with different tools and domains.***

# Where Agents Sit in the AI Landscape

*The bridge from Modules 1–4 (passive generation) to Module 5+ (autonomous action)*

*fundamental*

**Modules 1-2**

Understand & Prompt LLMs

Text generation, prompt engineering

**No agency**

**Modules 3-4**

LLM Architecture & Multimodal

Transformers, vision, audio, RAG basics

**No agency**

**Module 5**

Agentic AI

Perception, reasoning, tools, memory, planning

**Full agency**

**Module 9**

LLMOps & Production

Deploy, monitor, scale agent systems

**Production agency**

**Module 5 is the inflection point: LLMs stop being text generators and become autonomous problem solvers.**

# Key Takeaways

*The essential points from this concept*

- An AI agent is a goal-directed system** with a perceive→reason→act→observe loop — not a chatbot.
- Five capabilities distinguish agents:** perception, state, reasoning, tool use, and autonomous iteration.
- The spectrum runs from Level 0 (raw LLMs) to Level 4 (multi-agent)** — each adds capability.
- Real-world agents are deployed at scale:** Devin (coding), Klarna (support), Elicit (research), Claude Code (terminal).
- Standard LLMs are stateless, tool-less, single-turn** — agents solve all three.

**Next:** AI-AG-FN-TH-000002 — Agent Anatomy: Perception, Decision, Action



AI-AG-FN-TH-000001 Complete

# What is an AI Agent?

Concept 1 of 8

Advanced Certification in Agentic & Generative AI · IISc Bangalore

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